MIL-C-24643/30B
3 October 1986
SUPERSEDING
MIL-C-0024643/30A(SH)
8 November 1985
MIL-C-24643/30
28 September 1984

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LS1SWU (INCLUDING VARIATION LS1SWA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the cable described herein shall consist of this specification and the latest issue of MIL-C-24643.

This cable can be manufactured in two variations: LS1SWU (unarmored) and LS1SWA (armored).

REQUIREMENTS:

Qualification required.

Construction (watertight)

First - Copper conductor, in accordance with ASTM B 286, size 22-7, uncoated. Maximum diameter 0.033 inch.
Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third - Cross-linked polyethylene insulation. Nominal thickness of 0.048 inch; wall thickness may vary from nominal as necessary, in order for completed cable to meet the specified electrical requirements.
Fourth - Braided shield of AWG no. 36 uncoated copper. Braid angle of 30 to 35 degrees; minimum coverage of 85 percent.
Fifth - Shield insulation of two polyester tapes, type G in accordance with MIL-I-631, sealed. Standard identification code applied by method 2, on inner tape.
Sixth - The required number of singles (see table I) cabled with a lay not greater than 24 times the pitch diameter of the layer. Cabling sequence shall be consecutive, starting with no. 1, from the center outward. Fillers shall be used to form a well-rounded cable.
Seventh - Binder tape applied helically with overlap.

(B) denotes changes.

AMSC N/A FSC 6145
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Construction variant for type LS1SWU

Eighth - Cross-linked polyolefin jacket. (See table I for thickness.) Cable surface marking required.

Construction variant for type LS1SWA

Eighth - Cross-linked polyolefin jacket. (See table I for thickness.)
Seventh - Braided metal armor and paint.

TABLE I. Details.

Military 1/ part no. M24643/30	Size 2/	Number of conductors	Cable jacket thickness min (inch)	Overall diameter (inches)			
				Type LS1SWU		Type LS1SWA	
				min	max	min	max
-01UN	2	2	0.030	0.430	0.455	0.480	0.505
-02UN	14	14	.040	.825	.870	.875	.920
-03UN	20	20	.040	.970	1.030	1.020	1.080
-04UN	30	30	.050	1.135	1.200	1.185	1.250

1/ Unarmored. When armor option is required, see MIL-C-24643 for configuration of part number.
2/ Type LS1SWU and LS1SWA.

EXAMINATION AND TESTS:

	Requirements
Basic electrical:	
Conductor resistance - ohms/1000 feet, at 25°C, maximum.	17.2
Voltage withstand - volts, root mean square, minimum	
Conductor to shield.....	3000
Shield to shield.....	500
Insulation resistance - megohms/1000 feet, minimum.....	500
Conductor continuity.....	No failure
Group A:	
Visual and dimensional.....	No failure
Hydrostatic (open end) - leakage at 300 lb/in^2, 2 hours, in^3, maximum.....	1.0
Capacitance - 1 kilohertz (kHz), pF/ft, maximum.....	28
Characteristic impedance - at 1 MHz, ohms.....	61 ± 3

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EXAMINATION AND TESTS: (Continued)

	Requirements
Group B:	
Drip - 95 ± 1°C	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)	35
Physicals (unaged)	
Insulation	
Tensile strength - lb/in^2, minimum	1800
Elongation - percent, minimum	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Group C:	
Physicals (aged)	
Insulation	
Air oven	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged, minimum	80
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged, minimum	60
Hot oil immersion	
Tensile strength - lb/in^2, minimum	50
Elongation - percent, minimum	50
Shrinkage	No failure
Heat distortion - percent of unaged, maximum	30
Permanence of printing (jacket) - (for type LS1SWU only) - cycles, minimum	250
Cable filler removability	No failure
Armor (type LS1SWA only) - conformance to material, construction and coverage	No failure
Shield - conformance to material, construction and coverage	No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B and C, plus the following:

Aging and compatibility (cable) (125 ± 5°C)	No failure
(B) Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Flame propagation (cable)	No failure

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QUALIFICATION INSPECTION: (Continued)

	Requirements
Immersion (jacket)	
Tensile strength - percent of unaged, minimum.....	50
Elongation - percent of unaged, minimum.....	50
Smoke index, maximum	
Jacket.....	25
Fillers.....	--
(B) Insulation.....	45
Toxicity index, maximum	
Jacket.....	5
Fillers.....	5
Insulation.....	1.5
Durometer (jacket) - (type A) hardness, minimum.....	80
Weathering (jacket).....	No failure

UNIT ORDERING LENGTHS:
All sizes 1000 feet (nominal)

NOTE: Not for Air Force use.

Custodians:
Army - MI
Navy - SH

Review activities:
Army - AV, CR, ER
Navy - EC
DLA - IS

User activities:
Army - ME, AR, AL
Navy - CG

Preparing activity:
Navy - SH
(Project 6145-1005-30)

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